

Brutal Series - Paper B36 (Class 7: Light | Integers | Transportation in Animals & Plants | Symmetry)

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

1. A girl stands 2.4 m in front of a flat mirror, then walks 0.9 m straight toward it. What is the distance between the girl and her image now? [\[Light\]](#)
 - (A) 3.3 m
 - (B) 1.8 m
 - (C) 4.8 m
 - (D) 1.5 m
 - (E) 3.0 m
 - (F) 0.9 m

2. A submarine is at a depth of 180 m below sea level. It dives a further 45 m, then rises 96 m. At what depth is it now? [\[Integers\]](#)
 - (A) 321 m above sea level
 - (B) 231 m below sea level
 - (C) 129 m above sea level
 - (D) 129 m below sea level
 - (E) 321 m below sea level
 - (F) 39 m below sea level

3. A healthy adult's heart beats about 72 times each minute. About how many times does it beat in one full day? [\[Transportation in Animals & Plants\]](#)
 - (A) 1,03,680
 - (B) 1,728
 - (C) 10,36,800
 - (D) 4,320
 - (E) 43,200
 - (F) 17,280
 - (G) 10,368

4. A regular octagon has how many lines of symmetry, and what is its smallest angle of rotational symmetry? [\[Symmetry\]](#)
 - (A) 16 lines; 45 degrees
 - (B) 8 lines; 22.5 degrees
 - (C) 4 lines; 90 degrees
 - (D) 8 lines; 60 degrees
 - (E) 6 lines; 45 degrees
 - (F) 8 lines; 45 degrees
 - (G) 8 lines; 40 degrees

5. Two plane mirrors are placed facing each other at an angle of 45 degrees. How many images of a small object placed between them are seen? [\[Light\]](#)
- (A) 6
 - (B) 5
 - (C) 7
 - (D) 9
 - (E) 15
 - (F) 8
 - (G) 4
6. At 6 a.m. the temperature was -4 degrees C. It then rose by 3 degrees C every hour. What was the temperature at 12 noon? [\[Integers\]](#)
- (A) 11 degrees C
 - (B) 2 degrees C
 - (C) -22 degrees C
 - (D) 18 degrees C
 - (E) 14 degrees C
 - (F) -14 degrees C
 - (G) 21 degrees C
7. Timing for only 15 seconds, a health worker records 18 throbs at the wrist. What is the pulse rate per minute? [\[Transportation in Animals & Plants\]](#)
- (A) 270 beats per minute
 - (B) 18 beats per minute
 - (C) 36 beats per minute
 - (D) 90 beats per minute
 - (E) 108 beats per minute
 - (F) 60 beats per minute
 - (G) 72 beats per minute
8. Which capital letter has NO line of symmetry but HAS rotational symmetry of order 2? [\[Symmetry\]](#)
- (A) X
 - (B) H
 - (C) T
 - (D) S
 - (E) A
 - (F) B
 - (G) M
9. Between two plane mirrors a candle forms exactly 11 images. What is the angle between the mirrors? [\[Light\]](#)
- (A) 32.7 degrees
 - (B) 60 degrees
 - (C) 30 degrees
 - (D) 11 degrees
 - (E) 36 degrees
 - (F) 45 degrees
 - (G) 40 degrees

10. Simplify: $(-6) \times (-4) + (-8) \div 2 - (-5)$. [\[Integers\]](#)

- (A) 15
- (B) 25
- (C) 27
- (D) 23
- (E) 19
- (F) -25
- (G) 33

11. Which blood vessel carries oxygen-rich blood from the lungs back to the heart, breaking the usual rule for its type? [\[Transportation in Animals & Plants\]](#)

- (A) A capillary
- (B) Vena cava
- (C) Pulmonary vein
- (D) Aorta
- (E) Hepatic vein
- (F) Pulmonary artery

12. A shape looks exactly the same after turning through 72 degrees. What is the order of its rotational symmetry? [\[Symmetry\]](#)

- (A) 3
- (B) 4
- (C) 5
- (D) 10
- (E) 72
- (F) 6
- (G) 2

13. Light travels at about 3,00,000 km per second. How long does sunlight take to cross a distance of 18,00,000 km? [\[Light\]](#)

- (A) 60 s
- (B) 0.6 s
- (C) 18 s
- (D) 6 s
- (E) 5 s
- (F) 16 s
- (G) 3 s

14. Find the product: $(-2) \times (-3) \times (-4) \times (-5)$. [\[Integers\]](#)

- (A) 60
- (B) 14
- (C) -14
- (D) -120
- (E) 120
- (F) -60
- (G) 240

15. The human heart has four chambers and a fish heart has two. How many more chambers does the human heart have? [\[Transportation in Animals & Plants\]](#)

- (A) 1
- (B) 8
- (C) 0
- (D) 6
- (E) 4
- (F) 3
- (G) 2

16. What is the sum of the number of lines of symmetry of a square and of an equilateral triangle? [\[Symmetry\]](#)

- (A) 8
- (B) 12
- (C) 9
- (D) 5
- (E) 7
- (F) 4
- (G) 6

17. A vehicle's rear-view mirror always gives an erect, smaller image with a wide field of view. Which mirror is it, and why? [\[Light\]](#)

- (A) Concave mirror, because it narrows the view
- (B) Plane mirror, because it gives a wider field
- (C) Convex mirror, because it magnifies distant cars
- (D) Convex mirror, because it always gives a small, erect image and a wide view
- (E) Concave mirror, because it magnifies the image
- (F) Plane mirror, because the image is the same size

18. In a quiz, +4 marks are given for a correct answer and -2 for a wrong one. A student attempts all 25 questions and scores 58. How many did she answer correctly? [\[Integers\]](#)

- (A) 15
- (B) 14
- (C) 22
- (D) 12
- (E) 16
- (F) 20
- (G) 18

19. Red blood cells look red because they contain a red pigment that carries oxygen. Name it. [\[Transportation in Animals & Plants\]](#)

- (A) Haemoglobin
- (B) Melanin
- (C) Insulin
- (D) Chlorophyll
- (E) Plasma
- (F) A platelet
- (G) Carotene

20. A regular polygon has exactly 9 lines of symmetry. What is the measure of each of its interior angles?

[Symmetry]

- (A) 150 degrees
- (B) 135 degrees
- (C) 40 degrees
- (D) 160 degrees
- (E) 140 degrees
- (F) 144 degrees
- (G) 120 degrees

21. White sunlight splits into colours in a rainbow. A student lists Red, Orange, Yellow, Green, Blue and Violet. Which colour did the student miss? [Light]

- (A) Indigo and Pink
- (B) Pink
- (C) Indigo
- (D) White
- (E) Black
- (F) None, the list is complete
- (G) Brown

22. What is the sum of all integers from -15 to 12, both included? [Integers]

- (A) -27
- (B) -3
- (C) -45
- (D) -30
- (E) 3
- (F) -42
- (G) 42

23. Water rises in tall trees through narrow tubes. Which tissue carries water and minerals from the roots up to the leaves? [Transportation in Animals & Plants]

- (A) Veins
- (B) Root hairs
- (C) Stomata
- (D) Arteries
- (E) Cambium
- (F) Phloem
- (G) Xylem

24. An equilateral triangle is rotated about its centre. After how many degrees of turn does it first look exactly the same? [Symmetry]

- (A) 60 degrees
- (B) 3 degrees
- (C) 180 degrees
- (D) 360 degrees
- (E) 240 degrees
- (F) 120 degrees

- 25.** On an ambulance the word is printed reversed so a mirror shows it correctly. In the word AMBULANCE, how many letters look unchanged in a plane mirror (same after a left-right flip)? [\[Light\]](#)
- (A) 6
 - (B) 5
 - (C) 3
 - (D) 7
 - (E) 4
 - (F) 9
 - (G) 2
- 26.** The product of two integers is -48. If one of them is -6, what is the other? [\[Integers\]](#)
- (A) -54
 - (B) -42
 - (C) 8
 - (D) 288
 - (E) -8
 - (F) 42
- 27.** Food made in the leaves is carried to all parts of a plant by which tissue? [\[Transportation in Animals & Plants\]](#)
- (A) Arteries
 - (B) Xylem
 - (C) Roots
 - (D) Phloem
 - (E) Capillaries
 - (F) Stomata
 - (G) Bark
- 28.** Which figure has rotational symmetry of order more than 1 but NO line of symmetry? [\[Symmetry\]](#)
- (A) An equilateral triangle
 - (B) An isosceles triangle
 - (C) A square
 - (D) A parallelogram (not a rectangle or rhombus)
 - (E) A circle
 - (F) A rectangle
- 29.** An 18 cm tall pencil stands upright 25 cm from a plane mirror. What is the straight-line distance between the top of the pencil and the top of its image? [\[Light\]](#)
- (A) 11 cm
 - (B) 50 cm
 - (C) 25 cm
 - (D) 43 cm
 - (E) 36 cm
 - (F) 68 cm

- 30.** A lift starts 3 floors below ground (-3). It goes up 9, down 4, up 7, then down 12. Which floor does it stop at? [\[Integers\]](#)
- (A) 3rd floor below ground
 - (B) 7th floor below ground
 - (C) 5th floor below ground
 - (D) ground floor
 - (E) 1st floor below ground
 - (F) 3rd floor above ground
 - (G) 9th floor above ground
- 31.** A plant loses about 4 litres of water per hour by transpiration on a hot day. How much water does it lose in 6 hours? [\[Transportation in Animals & Plants\]](#)
- (A) $\frac{2}{3}$ litre
 - (B) 240 litres
 - (C) 18 litres
 - (D) 24 litres
 - (E) 12 litres
 - (F) 1.5 litres
 - (G) 10 litres
- 32.** The capital letter H has how many lines of symmetry and what order of rotational symmetry? [\[Symmetry\]](#)
- (A) 4 lines; order 4
 - (B) 2 lines; order 4
 - (C) 2 lines; order 2
 - (D) 2 lines; order 1
 - (E) 1 line; order 2
 - (F) 0 lines; order 2
 - (G) 1 line; order 1
- 33.** A periscope in a submarine uses two plane mirrors. At what angle is each mirror fixed so light turns by 90 degrees twice? [\[Light\]](#)
- (A) 22.5 degrees
 - (B) 30 degrees
 - (C) 45 degrees
 - (D) 180 degrees
 - (E) 90 degrees
 - (F) 40 degrees
 - (G) 60 degrees
- 34.** Evaluate: $(-100) \div (-4) \div 5 \times (-2)$. [\[Integers\]](#)
- (A) 2
 - (B) -2
 - (C) -250
 - (D) 10
 - (E) 50
 - (F) -10

35. Tiny pores on the underside of leaves through which water vapour escapes are called: [\[Transportation in Animals & Plants\]](#)

- (A) Lenticels
- (B) Cambium
- (C) Root hairs
- (D) Xylem
- (E) Stomata
- (F) Veins
- (G) Capillaries

36. A figure has rotational symmetry of order 6. Through which of these angles will it NOT look the same? [\[Symmetry\]](#)

- (A) 300 degrees
- (B) 180 degrees
- (C) 120 degrees
- (D) 240 degrees
- (E) 360 degrees
- (F) 90 degrees
- (G) 60 degrees

37. A doctor uses a mirror that gives a large, erect image of a tooth when held close to it. Which mirror is this? [\[Light\]](#)

- (A) Curved glass slab
- (B) Plane mirror
- (C) Convex mirror
- (D) Flat glass sheet
- (E) Concave lens
- (F) Convex lens
- (G) Concave mirror

38. The sum of two integers is -8. If one integer is 15, what is the other? [\[Integers\]](#)

- (A) -7
- (B) 23
- (C) 7
- (D) -30
- (E) -23
- (F) 13
- (G) -13

39. Which part of the blood helps it clot and stop bleeding from a cut? [\[Transportation in Animals & Plants\]](#)

- (A) Lymph
- (B) Haemoglobin
- (C) Red blood cells
- (D) White blood cells
- (E) Plasma
- (F) Platelets
- (G) Serum

40. How many lines of symmetry does a regular hexagon have? [\[Symmetry\]](#)

- (A) 4
- (B) 2
- (C) 8
- (D) 1
- (E) 12
- (F) 3
- (G) 6

41. A ray of light hits a plane mirror making an angle of 35 degrees with the mirror surface. What is the angle of reflection, measured from the normal? [\[Light\]](#)

- (A) 90 degrees
- (B) 70 degrees
- (C) 125 degrees
- (D) 55 degrees
- (E) 45 degrees
- (F) 35 degrees
- (G) 110 degrees

42. A test has 20 questions: +5 for correct, -3 for wrong, 0 for unattempted. Riya got 11 correct and 6 wrong. What is her score? [\[Integers\]](#)

- (A) 31
- (B) 19
- (C) 37
- (D) 49
- (E) 55
- (F) 73
- (G) 43

43. White blood cells are important mainly because they: [\[Transportation in Animals & Plants\]](#)

- (A) transport water in the body
- (B) give blood its red colour
- (C) carry oxygen
- (D) help blood to clot
- (E) carry food to cells
- (F) carry carbon dioxide away
- (G) fight germs and protect against disease

44. A photo frame is an ordinary rectangle, longer than it is wide. How many lines of symmetry does it have? [\[Symmetry\]](#)

- (A) 0
- (B) 8
- (C) 6
- (D) 3
- (E) 1
- (F) 4
- (G) 2

45. Two plane mirrors stand at 90 degrees to each other. How many images of a coin placed between them are formed? [\[Light\]](#)

- (A) 90
- (B) 6
- (C) 2
- (D) 5
- (E) 1
- (F) 3
- (G) 4

46. Which value is the greatest: $-|-7|$, $-(-5)$, $|-3| - |-9|$, or $(-2) \times (-4)$? [\[Integers\]](#)

- (A) $-(-5) = 5$
- (B) $-(-5) = -5$
- (C) $|-3| - |-9| = -6$
- (D) $-|-7| = 7$
- (E) $(-2) \times (-4) = -8$
- (F) $(-2) \times (-4) = 8$

47. The liquid part of blood that carries cells, food and wastes is called: [\[Transportation in Animals & Plants\]](#)

- (A) Mucus
- (B) Serum
- (C) Lymph
- (D) Platelets
- (E) Plasma
- (F) Haemoglobin
- (G) Cytoplasm

48. A perfectly round coin has the shape of a circle. In all, how many lines of symmetry does such a circle have? [\[Symmetry\]](#)

- (A) 4
- (B) 8
- (C) 1
- (D) Infinitely many
- (E) 360
- (F) none
- (G) 2

49. A clock seen in a plane mirror shows the time as 8:20. What is the actual time? [\[Light\]](#)

- (A) 9:40
- (B) 8:40
- (C) 3:20
- (D) 4:40
- (E) 3:40
- (F) 8:20

50. The temperature dropped from 7 degrees C to -11 degrees C overnight. By how many degrees did it fall? [\[Integers\]](#)

- (A) 11 degrees C
- (B) 22 degrees C
- (C) -4 degrees C
- (D) -18 degrees C
- (E) 7 degrees C
- (F) 18 degrees C

51. Blood is cleaned of wastes such as urea by two bean-shaped organs. They are the: [\[Transportation in Animals & Plants\]](#)

- (A) Stomach
- (B) Heart
- (C) Spleen
- (D) Lungs
- (E) Kidneys
- (F) Bladder

52. An equilateral triangle has 3 lines of symmetry and rotational symmetry of order 3. What is the product of these two numbers? [\[Symmetry\]](#)

- (A) 27
- (B) 6
- (C) 0
- (D) 12
- (E) 9
- (F) 1

53. In 5 seconds, how far does light travel if its speed is 3,00,000 km per second? Give the answer in km. [\[Light\]](#)

- (A) 30,00,000 km
- (B) 75,000 km
- (C) 3,00,000 km
- (D) 15,00,000 km
- (E) 60,000 km
- (F) 15,000 km

54. Evaluate: $-3 - [-4 + (6 - 9)] \times 2$. [\[Integers\]](#)

- (A) 17
- (B) 11
- (C) -17
- (D) -5
- (E) 5
- (F) -1

55. When a person's kidneys fail, their blood is cleaned by a machine in a process called: [\[Transportation in Animals & Plants\]](#)

- (A) Transfusion
- (B) Osmosis
- (C) Dialysis
- (D) Photosynthesis
- (E) Filtration in the liver
- (F) Excretion by skin

56. The smallest angle of rotational symmetry of a regular polygon is 30 degrees. How many sides does it have? [\[Symmetry\]](#)

- (A) 30
- (B) 24
- (C) 15
- (D) 6
- (E) 12
- (F) 10
- (G) 8

57. Light travels in straight lines, so a 2 m pole casts a 3 m shadow while a tree casts an 18 m shadow at the same time. How tall is the tree? [\[Light\]](#)

- (A) 12 m
- (B) 24 m
- (C) 36 m
- (D) 18 m
- (E) 6 m
- (F) 27 m

58. The product of three integers is negative. Which statement is always true? [\[Integers\]](#)

- (A) An odd number of the factors are negative
- (B) All three are negative
- (C) An even number are negative
- (D) None of them are negative
- (E) All three are positive
- (F) Exactly two are negative

59. An adult body holds about 5 litres of blood. If 250 mL is donated, what fraction of the blood is given?

[\[Transportation in Animals & Plants\]](#)

- (A) $\frac{1}{2}$
- (B) $\frac{1}{4}$
- (C) $\frac{1}{25}$
- (D) $\frac{1}{5}$
- (E) $\frac{1}{50}$
- (F) $\frac{1}{20}$

60. Which word in capitals reads the same when reflected in a horizontal mirror placed below it (each letter must have a horizontal line of symmetry)? [\[Symmetry\]](#)

- (A) TIME
- (B) MATH
- (C) BOOK
- (D) HELP
- (E) SUN
- (F) FACE
- (G) CODE

61. Which order correctly lists the seven colours of white light from the colour bent least by a prism to the colour bent most? [\[Light\]](#)

- (A) Orange, Red, Yellow, Green, Blue, Indigo, Violet
- (B) Blue, Green, Red, Yellow, Orange, Indigo, Violet
- (C) Red, Orange, Yellow, Green, Blue, Indigo, Violet
- (D) Red, Orange, Yellow, Blue, Green, Indigo, Violet
- (E) Violet, Indigo, Blue, Green, Yellow, Orange, Red
- (F) Violet, Blue, Indigo, Green, Orange, Yellow, Red
- (G) Red, Yellow, Orange, Green, Blue, Indigo, Violet

62. A shopkeeper records profit as positive and loss as negative. Over 5 days the records were +120, -75, -40, +95, -60 rupees. What is the net result? [\[Integers\]](#)

- (A) a loss of 40 rupees
- (B) a profit of 40 rupees
- (C) a loss of 390 rupees
- (D) a loss of 175 rupees
- (E) a profit of 390 rupees
- (F) no profit and no loss
- (G) a profit of 175 rupees

63. Compared with veins, why must the walls of arteries be especially thick, elastic and muscular?

[\[Transportation in Animals & Plants\]](#)

- (A) have valves to stop backflow
- (B) carry blood slowly back to the heart
- (C) are the smallest blood vessels
- (D) carry only deoxygenated blood
- (E) filter wastes from the blood
- (F) carry blood pumped at high pressure away from the heart
- (G) store blood for emergencies

64. A regular polygon has a smallest angle of rotation equal to 24 degrees. What is the sum of all its interior angles? [\[Symmetry\]](#)

- (A) 2160 degrees
- (B) 24 degrees
- (C) 1800 degrees
- (D) 2700 degrees
- (E) 2340 degrees
- (F) 3600 degrees
- (G) 156 degrees

65. A convex lens is used as a magnifying glass to read tiny print. The image you see is: [\[Light\]](#)

- (A) erect, same size and real
- (B) inverted, smaller and real
- (C) inverted, enlarged and real
- (D) erect, enlarged and virtual
- (E) erect, smaller and virtual
- (F) inverted, enlarged and virtual
- (G) inverted, same size and virtual

66. Divide and simplify: $(-144) / (12 - 18)$. [\[Integers\]](#)

- (A) 12
- (B) -12
- (C) 24
- (D) -72
- (E) 8
- (F) -24
- (G) 72

67. Veins have valves inside them mainly to: [\[Transportation in Animals & Plants\]](#)

- (A) store extra blood
- (B) speed up the blood
- (C) mix arterial and venous blood
- (D) prevent the backward flow of blood
- (E) pump the blood forward
- (F) add oxygen to the blood
- (G) filter wastes from the blood

68. Which figure has BOTH a line of symmetry AND rotational symmetry of order 2? [\[Symmetry\]](#)

- (A) A parallelogram
- (B) The letter R
- (C) The letter P
- (D) A scalene triangle
- (E) An isosceles triangle
- (F) A rectangle
- (G) A general trapezium

69. A toy car sits 1.2 m from a plane mirror. The mirror is then pulled 0.5 m away from the still car. By how much does its image shift? [\[Light\]](#)

- (A) 2.4 m
- (B) 0.25 m
- (C) 1.7 m
- (D) 3.4 m
- (E) 2.0 m
- (F) 0.5 m
- (G) 1.0 m

70. By how much is -5 greater than -12? [\[Integers\]](#)

- (A) -7
- (B) -17
- (C) 17
- (D) 5
- (E) -5
- (F) -2
- (G) 7

71. A doctor records a patient's heartbeat as 75 beats per minute. How many times does the heart beat in 4 hours? [\[Transportation in Animals & Plants\]](#)

- (A) 1,80,000
- (B) 1,800
- (C) 18,000
- (D) 300
- (E) 4,500
- (F) 45,000

72. A starfish shape has 5 identical arms. What are the order and smallest angle of its rotational symmetry? [\[Symmetry\]](#)

- (A) Order 5; 72 degrees
- (B) Order 5; 60 degrees
- (C) Order 10; 36 degrees
- (D) Order 5; 90 degrees
- (E) Order 72; 5 degrees
- (F) Order 5; 45 degrees
- (G) Order 4; 90 degrees

73. Sunlight is white. When it passes through a glass prism it spreads into a band of colours. This splitting is called: [\[Light\]](#)

- (A) Diffusion
- (B) Dispersion
- (C) Total reflection
- (D) Refraction only
- (E) Reflection
- (F) Lateral inversion

74. Fill the blank to make it true: $(-8) \times \underline{\hspace{1cm}} = 56$. [\[Integers\]](#)

- (A) -8
- (B) -448
- (C) 8
- (D) 6
- (E) 7
- (F) -7

75. Besides cooling the body, sweating by the skin also helps to remove: [\[Transportation in Animals & Plants\]](#)

- (A) excess water and salts (wastes)
- (B) carbon dioxide only
- (C) blood cells
- (D) food
- (E) oxygen
- (F) platelets
- (G) haemoglobin

76. How many lines of symmetry does an isosceles triangle that is NOT equilateral have? [\[Symmetry\]](#)

- (A) 3
- (B) 2
- (C) 0
- (D) 1
- (E) 5
- (F) 6
- (G) 4

77. A torch bulb is placed at the focus of a concave mirror, as in a car headlight. The reflected light forms:

[\[Light\]](#)

- (A) a diverging beam
- (B) an inverted beam
- (C) a coloured beam
- (D) a magnified bright spot
- (E) no beam at all
- (F) a parallel beam of light
- (G) a converging beam

78. A bird is 24 m above a calm lake. Its reflection appears as far below the surface. Taking the surface as 0, what is the vertical distance between the bird and its reflection? [\[Integers\]](#)

- (A) 0 m
- (B) -24 m
- (C) 24 m
- (D) 12 m
- (E) -48 m
- (F) 48 m
- (G) 36 m

79. The throbbing of an artery felt at the wrist each time the heart beats is called the: [\[Transportation in Animals & Plants\]](#)

- (A) clot
- (B) Pulse
- (C) reflex
- (D) heartbeat sound
- (E) lymph flow
- (F) muscle spasm
- (G) blood pressure

80. A regular pentagon and a regular hexagon are placed side by side. What is the total number of their lines of symmetry? [\[Symmetry\]](#)

- (A) 12
- (B) 30
- (C) 8
- (D) 13
- (E) 10
- (F) 9
- (G) 11

81. The image formed by a plane mirror has four properties. Which set is fully correct? [\[Light\]](#)

- (A) Virtual, inverted, magnified, reversed
- (B) Real, erect, smaller, reversed
- (C) Real, inverted, same size, not reversed
- (D) Virtual, erect, same size, laterally inverted
- (E) Virtual, erect, magnified, reversed
- (F) Real, erect, same size, not reversed
- (G) Virtual, inverted, same size, reversed

82. Simplify: $(15 - 23) \times (7 - 11) \div (-2)$. [\[Integers\]](#)

- (A) 16
- (B) 8
- (C) -8
- (D) -16
- (E) -32
- (F) -64
- (G) 64

83. During heavy exercise the heart rate rises from 72 to 120 beats per minute. By what percentage did it increase? [\[Transportation in Animals & Plants\]](#)

- (A) 52%
- (B) 166%
- (C) 48%
- (D) about 67%
- (E) 40%
- (F) 33%
- (G) 60%

84. The order of rotational symmetry of a square is 4. Through which set of angles (after 0 degrees, within one full turn) does it map onto itself? [\[Symmetry\]](#)

- (A) 45, 90, 135 and 180 degrees
- (B) 90, 270 and 360 degrees only
- (C) 30, 60, 90 and 120 degrees
- (D) 60, 120, 180 and 240 degrees
- (E) 90, 180 and 270 degrees only
- (F) 180 and 360 degrees only
- (G) 90, 180, 270 and 360 degrees

- 85.** Two mirrors are inclined so that 5 images are formed. If the angle between them is now halved, how many images are formed? [\[Light\]](#)
- (A) 9
 - (B) 11
 - (C) 12
 - (D) 23
 - (E) 5
 - (F) 6
 - (G) 10
- 86.** The successor of -1 is added to the predecessor of -1. What is the result? [\[Integers\]](#)
- (A) -1
 - (B) 2
 - (C) 3
 - (D) 1
 - (E) -2
 - (F) 0
 - (G) -3
- 87.** Lymph is a body fluid that: [\[Transportation in Animals & Plants\]](#)
- (A) pumps blood around the body
 - (B) is the red part of blood
 - (C) carries oxygen only
 - (D) clots a wound shut
 - (E) is found only in plants
 - (F) bathes the cells and drains back into the blood, carrying white cells
- 88.** Which of these has the LARGEST number of lines of symmetry? [\[Symmetry\]](#)
- (A) A regular heptagon
 - (B) An equilateral triangle
 - (C) A regular hexagon
 - (D) A square
 - (E) A rectangle
 - (F) A regular octagon
 - (G) A regular pentagon
- 89.** Light from the Moon takes about 1.3 s to reach Earth. Taking light's speed as 3,00,000 km per second, roughly how far away is the Moon? [\[Light\]](#)
- (A) 30,00,000 km
 - (B) 3,90,000 km
 - (C) 13,00,000 km
 - (D) 4,30,000 km
 - (E) 2,30,769 km
 - (F) 39,000 km

90. Two integers differ by 9. The larger of them is -4. Find their product. [\[Integers\]](#)

- (A) 65
- (B) -65
- (C) -13
- (D) -52
- (E) -36
- (F) 52
- (G) 36

91. The largest artery of the body, which carries blood from the heart to the rest of the body, is the:

[\[Transportation in Animals & Plants\]](#)

- (A) a capillary
- (B) Aorta
- (C) coronary vein
- (D) pulmonary artery
- (E) jugular vein
- (F) pulmonary vein

92. A design looks the same after every 40-degree turn. Is this a valid rotational symmetry, and if so what is its order? [\[Symmetry\]](#)

- (A) No; 40 does not divide 360
- (B) Yes; order 36
- (C) Yes; order 9
- (D) Yes; order 40
- (E) Yes; order 8
- (F) Yes; order 4
- (G) No; only even orders are allowed

93. An object placed in front of a plane mirror is 30 cm from its own image. How far is the object from the mirror? [\[Light\]](#)

- (A) 7.5 cm
- (B) 10 cm
- (C) 30 cm
- (D) 20 cm
- (E) 60 cm
- (F) 45 cm
- (G) 15 cm

94. Evaluate: $|-6 - 4| - |3 - 8| + (-2)$. [\[Integers\]](#)

- (A) 7
- (B) 17
- (C) 3
- (D) -7
- (E) -3
- (F) 1

95. Root hairs help a plant absorb water faster mainly by: [\[Transportation in Animals & Plants\]](#)

- (A) pumping water like a heart
- (B) pushing water with valves
- (C) filtering the soil particles
- (D) storing water for dry days
- (E) making food for the plant
- (F) producing oxygen in the soil
- (G) greatly increasing the surface area for absorption

96. The minute hand of a clock turns from the 12 to the 3. Through what angle has it turned, and is that a symmetry rotation of a square? [\[Symmetry\]](#)

- (A) 270 degrees; yes
- (B) 180 degrees; yes
- (C) 90 degrees; no
- (D) 90 degrees; yes
- (E) 90 degrees; only for a rectangle
- (F) 60 degrees; yes
- (G) 30 degrees; no

97. Why are we able to see a non-luminous object such as a book? [\[Light\]](#)

- (A) Because it is transparent
- (B) It glows in the dark on its own
- (C) Because our eyes give out light
- (D) It absorbs all the light falling on it
- (E) It refracts light into separate rays
- (F) It produces its own light
- (G) It reflects light from a source into our eyes

98. A number is multiplied by -3 and then increased by 14, giving -7. What is the number? [\[Integers\]](#)

- (A) 21
- (B) 3
- (C) -7
- (D) -21
- (E) 9
- (F) 7

99. Blood circulates roughly once each minute in a resting adult. About how many times does it circulate in one hour? [\[Transportation in Animals & Plants\]](#)

- (A) about 60 times
- (B) 600 times
- (C) 3600 times
- (D) 70 times
- (E) 24 times
- (F) 7 times

100. Each exterior angle of a certain regular polygon is 36 degrees. How many lines of symmetry does the polygon have? [\[Symmetry\]](#)

- (A) 10
- (B) 5
- (C) 20
- (D) 6
- (E) 8
- (F) 12
- (G) 36